



Gary Wu 吳振銓

Senior Full-Stack Engineer · Industrial System Integration Architect

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PROFILE

Senior full-stack engineer with 5+ years specialising in industrial system integration and edge-to-cloud architecture. I integrate edge device data, monitoring workflows, AI inference, and cloud business systems into stable, maintainable, long-term-evolvable digital systems.

HIGHLIGHTS

15+ repositories integrated **3x** debug localisation **4x** faster deployment

100% eventual data sync **5+** years experience

CORE SKILLS

Frontend	React · Next.js · TypeScript · Redux · D3.js · Tailwind CSS · Electron
Backend & DB	Node.js · Python (FastAPI / Flask) · Strapi · PostgreSQL · MySQL · MongoDB
Industrial & Edge	OPC UA / PLC · SCADA · IPC · Edge Computing · IPFS
AI & Dev	YOLOv8 / v11 · OpenCV · OpenAI API · AI Agent · SDD
Security	ISO 27001:2022 ISMS · Internal Audit · Risk Assessment

EXPERIENCE

Senior Full-Stack Engineer

2020.10 – Present

- Led the solar panel recycling system across 15+ repositories — IPC core architecture and edge-to-cloud cross-system integration; mastered 90%+ of repositories for rapid cross-service diagnosis
- Connected SCADA, PLC (OPC UA), 5 IPC edge modules, YOLO AI, 3 Cloud APIs, and IPFS verifiable storage into a complete device-to-cloud data loop
- Introduced config-driven design for device versions 1.0–2.x, cutting redundant development ~75%; a version manager cut IPC software deployment from 1 hour to 15 minutes
- Rebuilt edge logging into 764 structured call sites (6 levels / 61 modules / 52 events), scheduled to the cloud and filtered live on SCADA — improving debug localisation ~3x
- Designed an Edge-to-Cloud sync (real-time upload + scheduled retry) achieving 100% eventual data-sync success and 99.9% availability; plus off-site cloud DR within 24h RPO
- Built end-to-end traceability across 7 key stations with 21 process timestamps per panel; recycling data and system logs retained in the cloud for at least 5 years
- Supported AI object recognition end-to-end — data preparation, training, deployment, and IPC integration; served as primary technical liaison with the equipment manufacturer
- Also contributed to National Museum of Taiwan Literature web systems (10+ repositories) and several digital-humanities platforms using D3.js / Leaflet / timelines / charts

Freelance Full-Stack Engineer

2024.10 – Present

- Built the Rock System (notary / land-administration platform) end-to-end as sole engineer — requirements, architecture, full-stack, permissions, and containerised delivery
- Unified case management, petty cash, reporting, permissions, and monitoring in one platform — ~2x faster filing and ~10x faster lookups; env-driven permissions support per-client rules

Full-Stack Engineer • LINE Chatbot Developer • IT Lecturer

2020.07 – 2020.12

- Built LINE chatbot automation; served as IT lecturer delivering technical courses; participated in full-stack development

Adjunct Assistant

2020.08 – 2020.10

- Assisted with academic research and technical support

SELECTED CASE STUDIES

Solar Panel Recycling System — Edge-to-Cloud Industrial Integration

Spanning 15+ repositories; config-driven design cut dev redundancy ~75%, and log observability made debug localisation ~3x faster.

[Edge-to-Cloud](#) · [IPC](#) · [SCADA](#) · [OPC UA](#) · [Python](#)

Rock System — Full-Stack Business Platform (0→1)

Sole engineer across the SDLC: fine-grained permissions, seat control, 2FA, and containerised delivery.

[React](#) · [Strapi](#) · [PostgreSQL](#) · [Docker](#)

Legacy System Performance Rescue

Located the real bottleneck (client network) and refactored transfer to cut frontend requests to 1/5–1/4.

[Performance](#) · [API Design](#) · [AI Collaboration](#)

Offline iPad Exhibition Setup

Self-hosted a local API on the iPad so video playback became fully independent of venue bandwidth.

[Edge](#) · [Alpine Linux](#) · [Python](#) · [React](#)

CERTIFICATIONS

ISO 27001:2022 ISMS

ISMS implementation, internal audit, risk assessment & treatment.

EDUCATION

National Yang Ming University

Bachelor's degree · Clinical Laboratory Science / Medical Technology

2001 – 2005